

Memory Portal: Investigating Data Meaning-making through Spatio-temporal Experiences

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Figure 1: Portal, an interactive mirror that reflects the shadows of past occupants of a space in the present.

ABSTRACT

The data collected in shared spaces are usually used to understand how occupants use the space. However, these data could also inform the experience of the occupants in these spaces over time. How could we leverage spatial and temporal memories as mediators among occupants of a shared physical space? We developed Memory Portal, an interactive mirror that reflects the shadows and

sounds of past occupants of a shared physical space in the present. We conducted a short user study to explore how the occupants of a shared space perceive the interactions with the reflected memories. One of our key finding is that the occupants formed a novel perspective about their actions and experience in the space through these memories. Our contributions include introducing a novel technical system and new approaches of data engagement in shared physical spaces through the visualization and sonification of memories of the presence of its occupants.



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CCS CONCEPTS

• **B7; Human-centered computing** → Interaction design; Interaction design process and methods; User centered design.

KEYWORDS

interaction design, embodied interaction, ubiquitous computing, smart environments, Connected Home

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1 INTRODUCTION

The data collected in shared public spaces are usually used to understand how occupants use the space [3]. However, these collected data could also shape the experience of the occupants in these spaces over time. In this paper, we investigate these data in the form of temporal and spatial memories in a physical shared space. How could we leverage spatial and temporal memories as mediators among occupants of a shared physical space? How do occupants of shared spaces make meaning of these collected data? How could a room or a shared space remember its occupants? we investigate new modes of data engagement in a shared physical space through reflecting memories of that space of its occupants. By applying a research-through-design methodology (RtD) [13], we designed Memory Portal, an interactive mirror that reflects the shadows and sounds of past occupants of a shared physical space in the present (Figure 1). The portal creates an intentional modification of temporality as a means of exploring how to make explicit the temporal connections between spatial and temporal memories, the portal, and those that were there before.

As an initial prototyping and experimentation step, we designed the Portal to reflect both the current and past occupants' shadows behind a wooden surface on the wall. We tested the prototype based on the Wizard of Oz method. We found that memories were clear when we added time and date stamps to each memory. Based on the initial prototype, we improved the Memory Portal and the interactions with it by including nonverbal audio memories along the visual abstractions of past occupants. We conducted a short study to investigate how the occupants of a shared space make meaning of visualized memories that show previous space occupants.

The Portal chooses memories of past occupants based on the number of current occupants in the physical shared space. While our investigation focused on a shared lounge for a design graduate cohort as a social space, the investigation has the potential to be applied in a shared residential setting such as student housing. Our contributions include introducing a novel technical system and new approaches of data engagement in shared physical spaces through the visualization and sonification of memories of the presence of its occupants.

2 RELATED WORK

2.1 Temporality and memory

Our minds understand time by what we can see, touch, hear, and taste. Design anthropologist Tim Ingold explains that we carry maps in our heads that blend our past observations and memories over the environment in the present [5]. He explains that the layering of past

spatial experiences with current ones in a physical environment amplifies these experiences.

Temporality is perhaps the most fundamental constituent of human cognition, but how it is perceived is different for each person based on the experiences that build their internal temporality [10]. Designers and HCI researchers have been investigating approaches for leveraging and implementing temporality in users' interactions [1, 2, 8, 9, 12]. Memories are something you cannot put your finger on, existing undefined at the edges of your consciousness. Through the Olo Radio, a music player that lets people re-experience digital music they have listened to previously, Odom et al. explored how any memory can be turned into a tangible experience. The radio creates a physical interaction with your memory as if you are putting a knob on your memory [9].

2.2 Embodied interactions for social engagement

Roquet and Sas explored how embodied interactions between the body and novel technologies can be used to create adaptive and generative spaces [3]. They explained that these spaces could guide sensory experiences to bring sensations in and out of focus. Other work demonstrated the importance of ambient displays to foster new types of interactions in public spaces. Vogel and Balakrishnan investigated enabling experiences in spaces that give people a choice to move between implicit and explicit interaction phases [11]. They developed a system for interacting and controlling the data present in their display.

Over the years, there have been several art installations that create experiences based on embodied interactions, such as; Myron Krueger's Videoplace [14], Daniel Rozin's Wooden Mirror [4], Camille Utterback's Liquid Time Series [5] and Entangled [6], Scott Snibbe's projects [7], and Klaus Obermaier's playful EGO [8], to name just a few. Scott Snibbe explained how visceral interactions could lead to social and emotional engagement [9]. His explorations of interactive art and tangible interfaces provided the foundation into the breadth of interactions possible for social engagement.

3 DESIGN QUESTIONS AND IMPLEMENTATION

We designed the Portal as an interactive interface mirror that reflects the shadows of past occupants of a shared physical space in the present. The portal allows for passive and active interactions with current and past time periods. Based on Roquet and Sas work [4], we designed the focal points that define the portal's interaction space to create experiences that build on each other and allow the occupants of the shared space to explore time as a part of the shared physical space.

The portal uses depth data from a Kinect to generate the participants' silhouettes and to determine when to move from a neutral ambient phase to one in which the participant is actively adding to the portal's "memories" using an LED matrix. We acknowledge that privacy and surveillance concerns might come to mind [6, 7]. For this reason and because of the study's aim, the portal shows the silhouette of the occupants without any indication of their identity. We chose the matrix because we wanted to create an interface that does not look like a monitor or a projection, but rather a part of

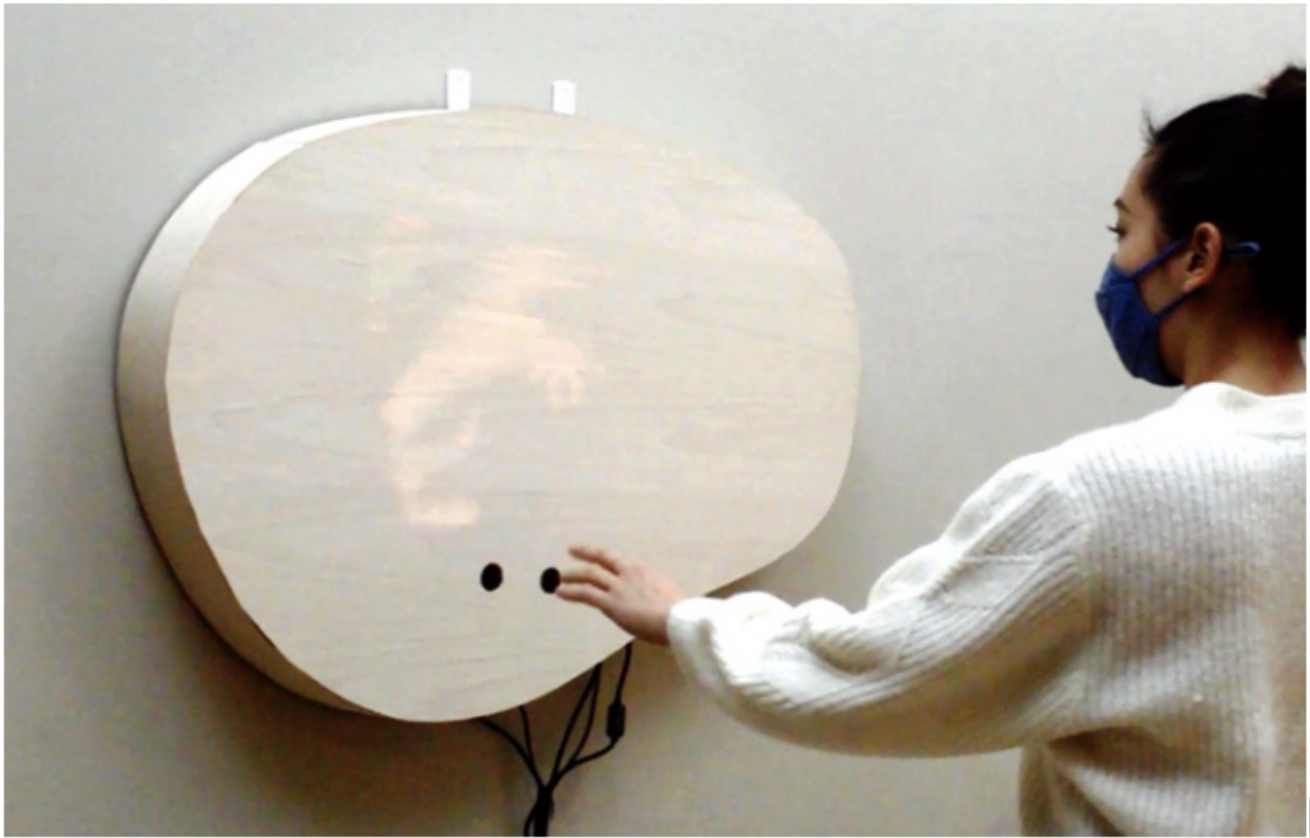


Figure 2: Early prototype of the Memory Portal. Here, a user is interacting with memories of space. The memories of occupants of the space shine through the wooden layer. The memories are produced from the RGB LED matrix attached to raspberry pi and a Kinect.

the physical space. So, we initially used wood veneer as a surface covering the matrix in the first phase.

We conducted one main experiment over one day in addition to initial exploration and having it running in our lab. We acknowledge that one day is a short time for the study. However, we were not able to extend the study due to covid-19 precautions. We plan on installing the portal for a longer time period in the future. In the beginning, we created an experiment based on The Wizard of Oz method. We asked the participants to interact with the portal for as long as they wanted and then conducted semi-structured interviews with them. Based on the insights from the first experiment, we created an updated iteration of the Portal. The design decisions will be extensively explained in the second phase section.

3.1 Context

We chose a shared lounge and workspace for grad students in a college's studio as a context for our experiments. The shared space consisted of a large kitchen, sofas, whiteboards, and tables for work. We chose that space because we wanted to have a large group of participants occupying a space regularly on a daily basis. Only 40 students inhabit the space.

3.2 Initial prototyping and experimentation

We designed the early prototype with an organic shape to create a contrast with screen-based interactions. We used wood as the main material, which led us to choose RGC-LED matrices to shine through the thin layer of veneer (see Figure 2). It hides the sensors and LEDs to only reflect the silhouette of the person in the memories. Our intention was that these abstractions of sound and visuals will be perceived as memories of the space. The Portal consisted of: (1) the Kinect to capture body position data, (2) the Raspberry Pi to process data, and (3) a set of RGB LED Matrices (Figure 3). We then placed it in the lounge. Although we had only the visual interaction working, we created an experiment based on the Wizard of Oz method to evaluate the other interactions.

3.2.1 Participants. We recruited 6 participants aged between 22-30 years old. We ran the study for one day. Four of the participants identified as female, and two identified as male. Each session included interaction with the portal for as long as they like and a semi-structured interview for about fifteen minutes.

3.2.2 Findings and synthesis. Regarding the concept and interactions, users liked the idea of having a “window” to interact with other spatial memories of the same space. They also preferred the

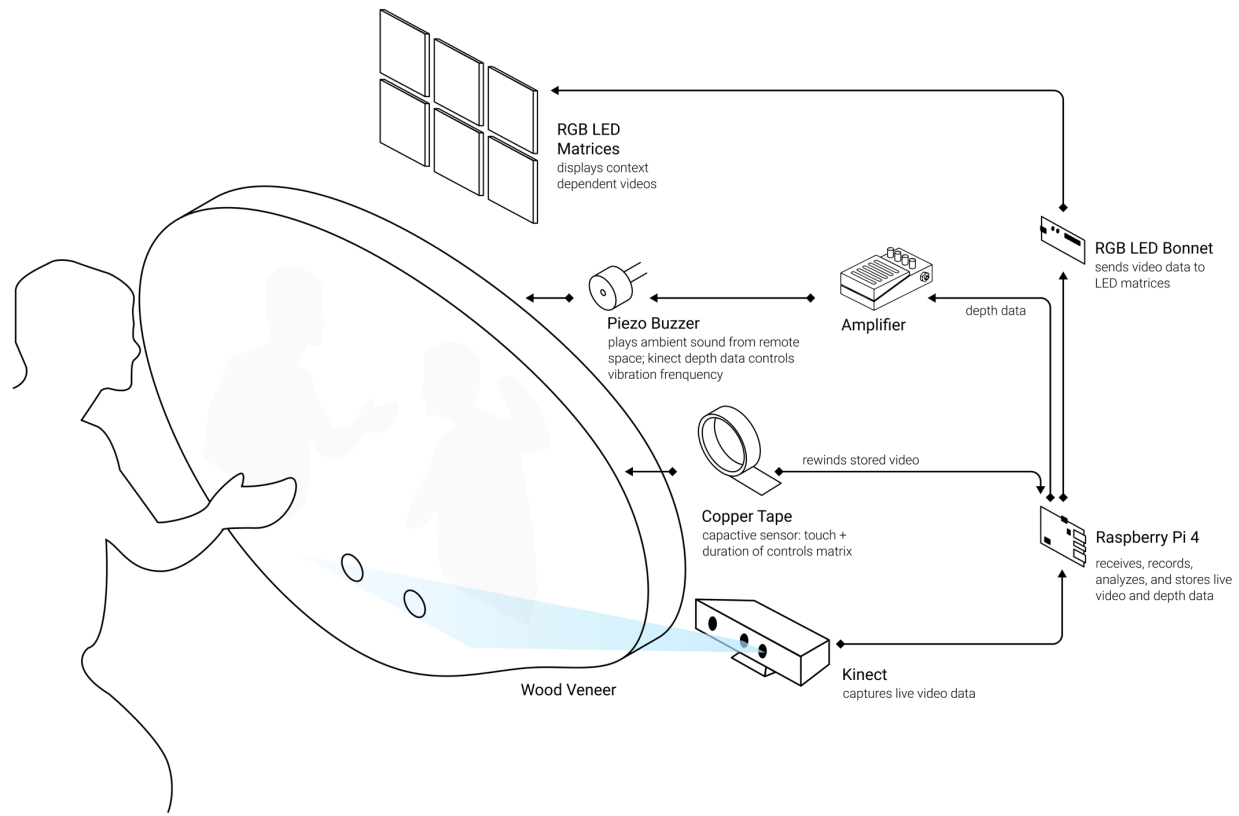


Figure 3: System diagram of the portal’s technical components showing the integration of several interactions such as visual, audio, and tactile.

anonymity of the visualization because it does not display the user’s real video and identity. They also favored the proximity-based interaction, in which as you get closer to the portal, the audio gets louder. When asked to react to the ghost reflection, participants stated that the ghost reflections are “a bit haunting, uncanny.” All participants indicated that they prefer more distinction between past and present reflections. The concept of memories or interacting with memories was difficult to grasp from seeing two overlapping moments in time, past and present. This led us in the second phase to visually add the date of the recorded memory.

3.3 Memory Portal

Based on the insights from the earlier exploration, we created an updated iteration of the Portal. We decided to keep a few features, such as the anonymity of the shadows that appear from the memories. To create a more significant distinction between the memory being played on the Portal and the current occupant of the space in the present, we used a mirror to reflect the presence of the person in real-time instead of the ghost reflection on top of the playing memory that confused participants (Figure 4). We also added the date to be shown on the top of each memory, which proved later its success in conveying the concept of the memory being played to the space occupants in the present (Figure 5).



Figure 4: The date of each memory is placed on the top of the portal. The current occupant’s image is reflected on the see-through mirror to avoid abstracting them into another digital shadow that overlaps with the memory.

We used two Raspberry Pi 4 B, one Kinect (for Xbox 360), and 12 32x32 RGB LED matrix panels. The main workflow can be described as follows: 1) the Kinect reads the silhouettes of people

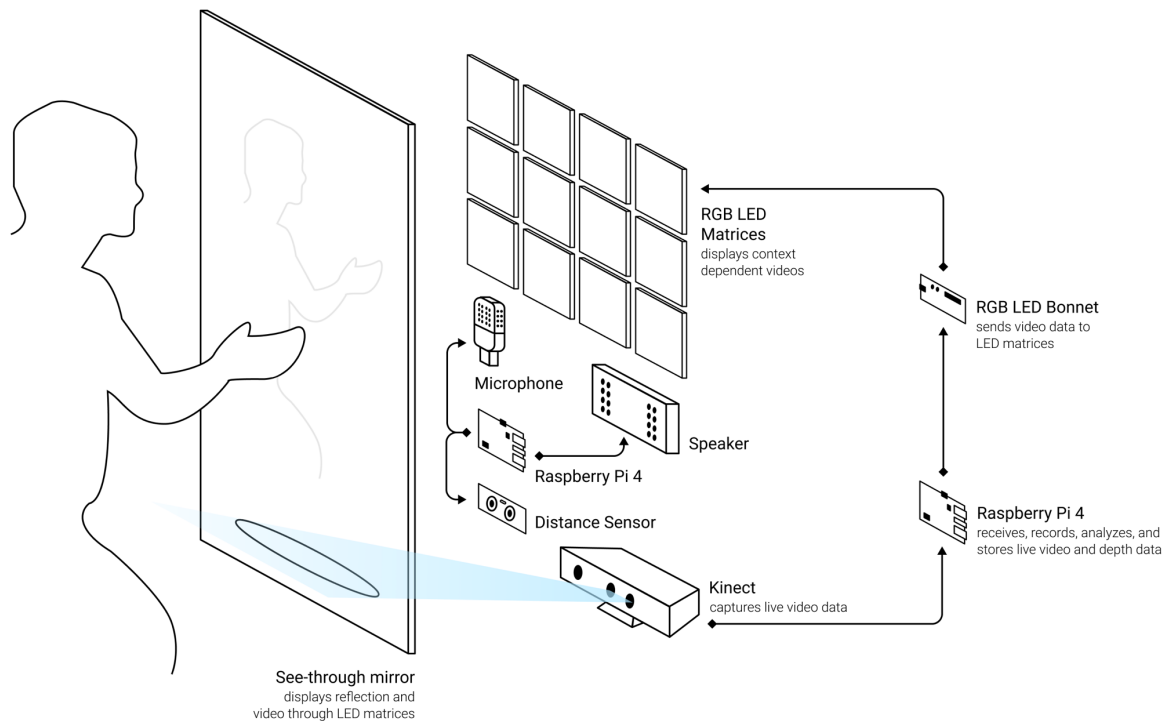


Figure 5: System diagram showing the see-through mirror and the RGB LED matrices behind it. The matrices are controlled by a pi that is connected to a Kinect to record the memories/silhouettes of occupants. Another Raspberry pi controls recording and playing the audio for each memory.

present and generates the depth frame; 2) the depth frames are then processed using OpenCV in one Raspberry Pi to produce video chunks. Considering the efficiency of reading and displaying the video we have captured in the past, it is more efficient to save the video in small chunks (30 fps and one minute long) instead of a large one (Figure 6); 3) the Raspberry Pi runs the face detection program (OpenCV Histogram of Oriented Gradients algorithm) to check the number of people present in each video frame which helps to decide where to save the final video file. This procedure is optional, in case some users want to see more/fewer people in the captured videos; 4) the saved videos are then sent/published to the other Raspberry Pi using ZeroMQ for stability and efficiency of the whole system, we separated functions into two Raspberry Pis; 5) the transferred videos are then processed to the format of a NumPy array with 1 byte per pixel, 3 channels RGB that can be displayed on the LED matrices.

We also updated the audio feature that changes in volume to higher or lower according to the proximity of current space occupants to and from the portal. The closer the occupant to the Portal, the louder the audio of the memory being played.

3.3.1 Less people/more people. We designed the portal to count people in the space and play a memory based on the number of people currently in the room. We created two scenarios, match, and mismatch. In match, it plays a reflection of the same number of people now occupying the space. A group of 4 or higher of current occupants= the Portal plays a memory of a large group of people. 1-3 occupants in the space= it plays a memory that has 1-3 people in it. In the mismatch scenario, if one person is detected in the space, the portal plays a memory for a large group (Figure 7). Later in the interviews, all participants indicated that the distinction was not clear to them; some preferred to see memories of larger groups of people independent of how many occupants are currently in the space.

3.3.2 Participants. We recruited 10 participants aged between 19-30 years old. We ran the study for one day. Most of the participants were grad students in the School of Design in Carnegie Mellon University except two undergraduate students who happened to be in the lounge area. Seven of the participants identified as female (she/her), two identified as male (he/him), and one identified as genderqueer (he/they).

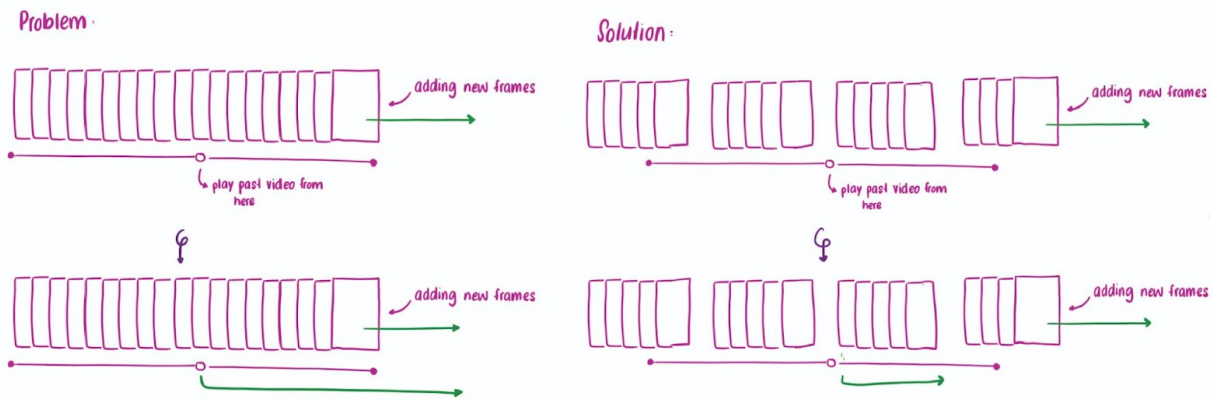


Figure 6: Video frame structure analysis. a) problem with reading and writing frames at the same time (memory consuming, slow process), and b) solution that saves video in small chunks and reads from previously saved ones (efficient).

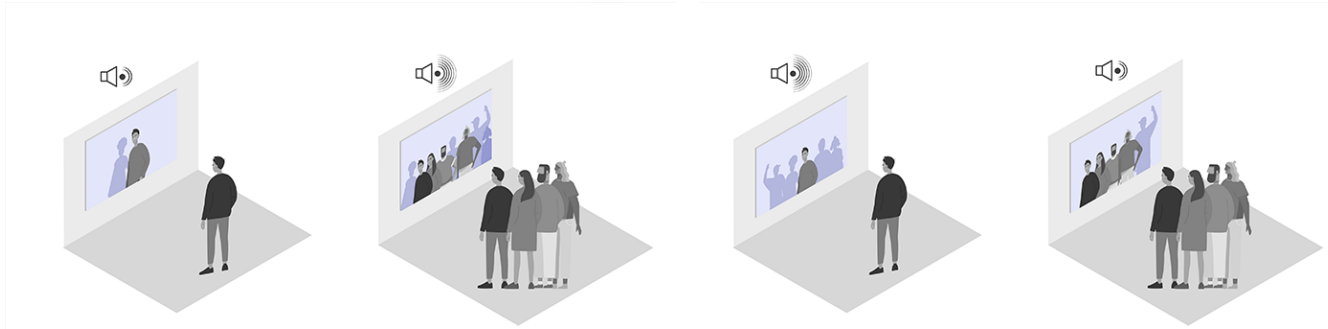


Figure 7: Match scenario, 1-3 people in the space = the Portal plays a memory of the same number of 1-3 people. Left: Mismatch: if one person is detected in the space, the portal plays a memory for a large group and vice versa.

3.3.3 Procedure. Each session included an interaction with the portal for as long as the participants wanted and a semi-structured interview for about fifteen minutes. We asked the participants to interact with the Portal as they see fit and just a think-aloud voicing their interactions and thoughts. We then conducted a semi-structured interview with each participant because we wanted to be open to novel suggestions for future development, including an evaluation of the intelligence of the system on a scale 1 to 5, in which 5 is the smartest since perceiving the interface as intelligent is a future step in our research. This was done to establish the areas of resonance and success of the Portal but also establish guidelines for creating interactions to achieve the perception of the memories within the space (Figure 8).

3.3.4 Findings and synthesis. On a scale 1-5, the participants evaluated the intelligence of the portal as 2.9 before we explained to them what the Portal does. After explaining it, the average score rose to 3.6 out of 5. However, most of the participants indicated that they want to experience it over more extended period of time. Due to the COVID-19 Pandemic and institutional restrictions, the duration of the study was limited. When asked to evaluate the Portal

after understanding more of the interactions, participants felt that they would need a timescale of weeks or months to truly evaluate the Portal's affordances. However, based on the interactions in the user study, participants wished to continue to interact and view the unfolding of the interactions over time, indicating they saw potential in the device's ability to grow or "learn" over time.

"I think it's showing us footage of based on some sort of interactions that we are having with it. It's quite ambient. The discovery of it kept unfolding as I interacted with it." — P5

"It's very magnetic. It really has my attention. There's not a lot out there in technology that responds to memory and spaces, and I feel like that is a very human thing." — P10

"Right now, I feel like there is a depth lacking in the interaction. I need more time to experience the device to develop a relationship over time." — P03

"This would be good in a studio context because there are times it's very busy and other times its empty and



Figure 8: Participants reacting to the portal and the memories shown on it.

then you could recognize yourself in the memories over a week or a month.” — P01

The scale and location of the Portal affected comfort and curiosity level of the interactions. When asked about possible improvement, several participants voiced a desire to expand the scale of the device to cover an entire wall or area of a communal space. However, they did not want to feel as if they were being observed by the Portal. People expressed a desire for a Portal in public spaces but expressed strong reservations about being placed in more private and intimate spaces.

“If the scale was larger, would it feel like memories of the people or would it feel like memories of the space.”
— P06

Audio played an important role in the interactions. When the Portal played memories without any audio, the participants tried to touch, move or interact with the portal in any way. It was unclear the level of interactivity of the Portal without the audio interaction of volume change depending on the proximity of the participants.

“There was no feedback, so I couldn’t tell if my interactions were affecting it. It’s like the interaction of “Am I on mute” because you are not sure if you are being interacted with it or not.” — P05

3.3.5 Discussion and future work. We found that through collected memories, the occupants of the shared space formed a novel perspective about their actions and experience in the space. A longer duration for the study is needed to gain a better perspective on how the Portal changes the occupants’ perception of the space. This gives them a chance to build a mental image about what the Portal does and its interaction patterns. Another point that we will take into consideration is adding more details in time stamps, maybe weeks as suggested. Audio is a strong feature in conveying the sense of memory of a communal space. The anonymity of the silhouettes of the occupants in the memories showed that it intrigued curiosity and engagement in knowing whose “shadow” is being reflected. The portal learning and growing as suggested by one of the participants aligns with our future step of treating each memory as data and growing the portal using ML as an intelligent interface.

4 CONCLUSION

In this research, we look at how a shared space that gets occupied by several recurring users remembers them and shares what it remembers. Our findings show users need to experience the interactions of these memories over a long duration of time. Due to the COVID-19 Pandemic and institutional restrictions, the course of the study was limited, and a more comprehensive study is needed to develop more results.

We conducted two studies to investigate how occupants of shared spaces make meaning of collected data when it is materialized in front of them. Our contributions include: (a) introducing new modes of data engagement in a shared physical space through materializing memories of the space of its occupants. (b) leveraging data as a mediator between users of space over time.

We chose a shared lounge for graduate students in a college as a context for our experiments. We installed Portal in the common lounge and asked the occupants to interact with it as they like. We conducted user interviews with the space occupants. The first study was partially based on the Wizard of oz method. Results revealed that the portal conveyed a sense of data collection when it reflected date-labeled memories of past occupants based on the number of current occupants. While our investigation focused on a common lounge as a social space, the results have the potential to be applied in shared spaces in residential settings, such a family game room or a graduate dorm.

Future iterations of the portal design envision gestural interactions to explicitly control the time and location of the spatial memories of that space. For future work, we would like to design similar experiments based solely on audio memories. The experiments show that audio is a vital feature in conveying the sense of memory of a space. We would like to develop more interactive audio-visual interactions with the current portal.

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